

ROLL NO :250701573

EXPERIMENT NO:10 EXPERIMENT TO UNDERSTAND K-MEANS CLUSTERING

ALGORITHM FOR A GIVEN DATASET

PROGRAM:

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
%matplotlib inline
```

```
df = pd.read_csv('Mall_Customers_modified.csv')
```

```
df.info()
```

```
df.head()
```

```
sns.pairplot(df)
```

```
features = df.iloc[:, [3, 4]].values
```

```
Features
```

```
from sklearn.cluster import KMeans
```

```
model = KMeans(n_clusters=5)
```

```
model.fit(features)
```

```
Final = df.iloc[:, [3, 4]].copy()
```

```
Final['label'] = model.predict(features)
```

```
Final.head()
```

```
sns.set_style("whitegrid")
```

```
sns.FacetGrid(Final, hue="label", height=8) \
.map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
.add_legend()
```

```
plt.show()
features_el = df.iloc[:, [2, 3, 4]].values
```

```
from sklearn.cluster import KMeans
```

```
wcss = []
```

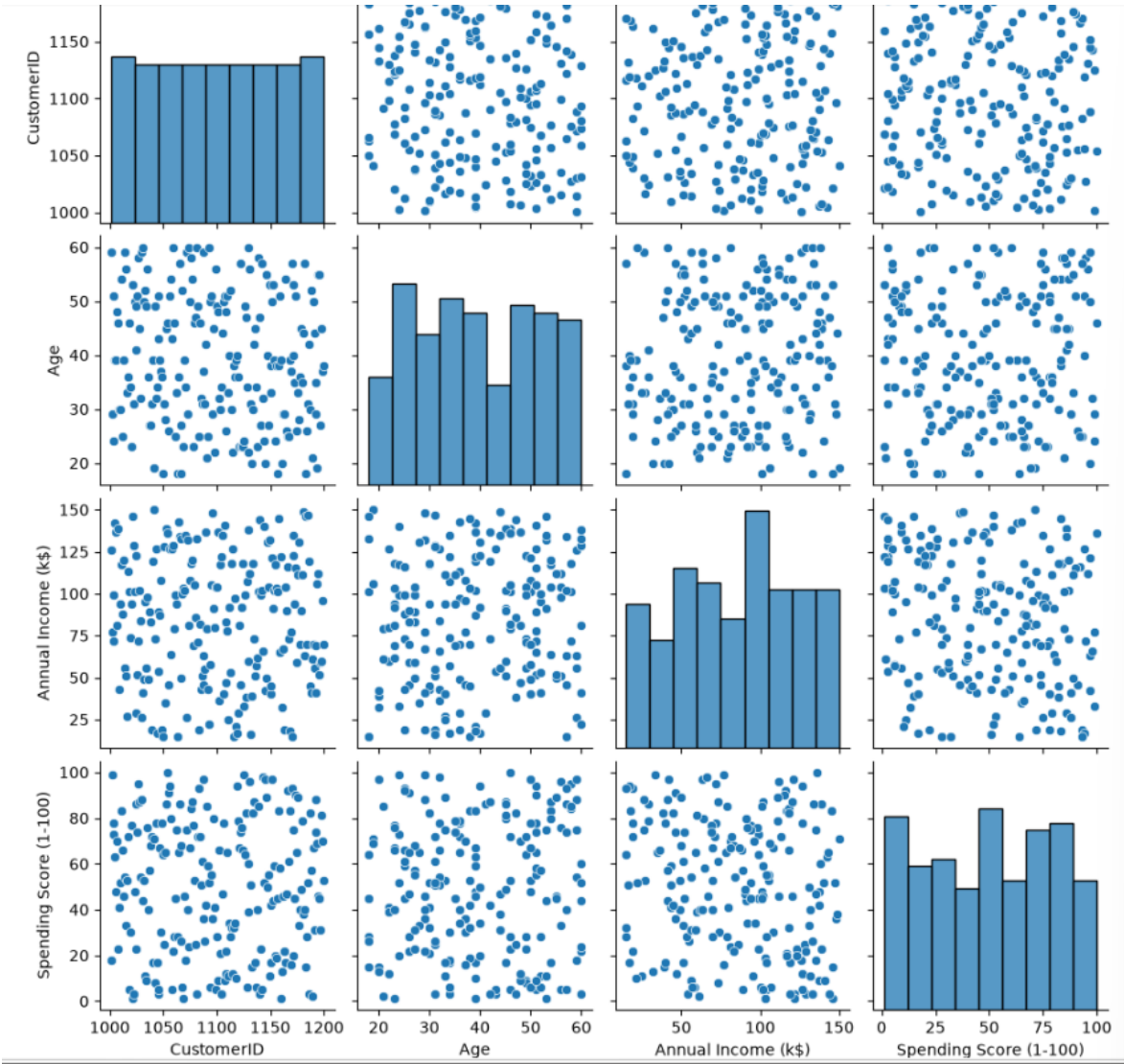
```
for i in range(1, 10):
    model = KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1, 10), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

OUTPUT:

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   str
2   Age                   200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), str(1)
```

memory usage: 7.9 KB

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1001	Female	59	126	18
1	1002	Male	29	77	99
2	1003	Female	24	99	73
3	1004	Male	51	72	78
					63
4	1005	Female	39	142	



```
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[96, 70], [70, 53]])				
Annual Income (k\$)	Spending Score (1-100)	label		
0	126	18		
1	77	99		
2	99	73		
3	72	78		
4	142	63		

